Ravi Kumar

Information Science & Engineering Student

Bengaluru, India | +91 77070 00206 | ravi7481081raj@gmail.com
github.com/ravimehta251 | linkedin.com/in/ravi-mehta2511 | leetcode.com/u/ravi_kumar_129

Education

Visvesvaraya Technological University (VTU) Belagavi, Karnataka
Bachelor of Engineering in Information Science and Engineering Sep 2023 – Jul 2027
CGPA: 8.1/10 | **Expected graduation:** July 2027
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems

Technical Skills

Languages: Java 21, SQL, JavaScript (ES6+), TypeScript
Backend: Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, Swagger/OpenAPI
Databases & Messaging: PostgreSQL, MySQL, Redis, Apache Kafka, pgvector
Testing: JUnit 5, Testcontainers, Spring Security Test, Postman
Cloud & Tools: Docker, AWS EC2, GitHub Actions (CI/CD), Git, Maven, Nginx, Flyway
Frontend & AI: React, Vite, Tailwind CSS, Spring AI, Azure OpenAI, Ollama, RAG

Projects

Bidly – Real-Time Auction Platform github.com/ravimehta251/bidly
Java 21 | Spring Boot 3.3 | PostgreSQL | Redis (Redisson) | React 19

- Combined Redisson distributed locks with JPA optimistic locking to protect concurrent bid writes; observed **zero conflicting writes in a 200-thread concurrency test**.
- Broadcast live price and auction-close events over **WebSocket (STOMP)** and Redis Pub/Sub across backend instances behind an Nginx load balancer.
- Built JWT-secured REST APIs for auction creation, browsing and bidding; tested backend behavior with **JUnit 5, Spring Security Test and Testcontainers**.

ShopMesh – Event-Driven E-Commerce github.com/ravimehta251/ShopMesh
Java 21 | Spring Boot 3.3 | Spring Cloud Gateway | Eureka | Kafka | MySQL

- Built **five Spring Boot services**: inventory, order and payment, plus an API gateway and service registry; orchestrated the stack with Docker Compose.
- Implemented a checkout saga using the **transactional outbox pattern and six Kafka event types**; added compensating events to release reserved stock when payment fails.
- Versioned **three independent MySQL databases** with Flyway and enforced stateless JWT authorization at the gateway.
- Automated deployment of the Docker Compose stack to **AWS EC2** on pushes to main using GitHub Actions.

Recall – Private Document Workspace github.com/ravimehta251/Recall
Java 21 | Spring Boot 3 | Spring AI | PostgreSQL 16 + pgvector | React 19 | TypeScript

- Built PDF, DOCX and TXT ingestion with Spring AI; chunked content, generated Azure OpenAI embeddings and stored them in PostgreSQL/pgvector for **retrieval-augmented generation (RAG)**.
- Streamed chat answers with source citations over authenticated **Server-Sent Events (SSE)**, emitting token, citation and completion events.
- Secured workspace, document and chat APIs with JWT and Spring Security; built a React/TypeScript UI and supported local Ollama models.

Achievements

- **2nd Runner-Up, JSSATE Hackathon:** Built Snap2Clean, a civic-issue reporting application using React, Google Maps API and MySQL.
- **Solved 500+ DSA problems on LeetCode** across arrays, strings, graphs and trees.